

Data Analytics (DA)- Project II

Analyzing E-Learning Platform Purchases using MySQL

Project Overview

This project analyzes purchase data from an online learning platform using MySQL. The database consists of three related tables: Learners, Courses, and Purchases. SQL queries were used to explore learner purchasing behaviour, identify popular courses and categories, calculate revenue, and generate business insights.

Key Findings

- Total spending was calculated for each learner to identify high-value customers.
- Analytics and Programming categories generated significant revenue compared to other categories.
- Some learners purchased courses from multiple categories, indicating strong engagement with the platform.
- The top-selling courses contributed a major portion of overall sales.
- A few courses showed little or no purchase activity, highlighting opportunities for promotional campaigns.
- Revenue and learner participation varied across course categories.

SQL Concepts Applied

- Database creation and table relationships using Primary and Foreign Keys.
- Data insertion and management.
- INNER JOIN, LEFT JOIN, and RIGHT JOIN operations.
- Aggregate functions such as SUM(), AVG(), COUNT(), and GROUP BY.
- Subqueries and correlated subqueries.
- Common Table Expressions (CTEs).
- CASE expressions for learner classification.
- NULL handling using COALESCE().
- View creation for category performance analysis.

Recommendations

1. Increase marketing efforts for high-performing course categories.
2. Create bundled course offers to encourage cross-category purchases.
3. Provide personalized recommendations to repeat learners.
4. Promote underperforming courses through discounts and targeted campaigns.
5. Focus on retaining high-value learners through loyalty programs and exclusive offers.

Conclusion

The analysis provides valuable insights into learner behavior, course popularity, and revenue performance. By leveraging SQL-based reporting and analytics, the e-learning platform can make data-driven decisions to improve sales, customer engagement, and overall business growth.